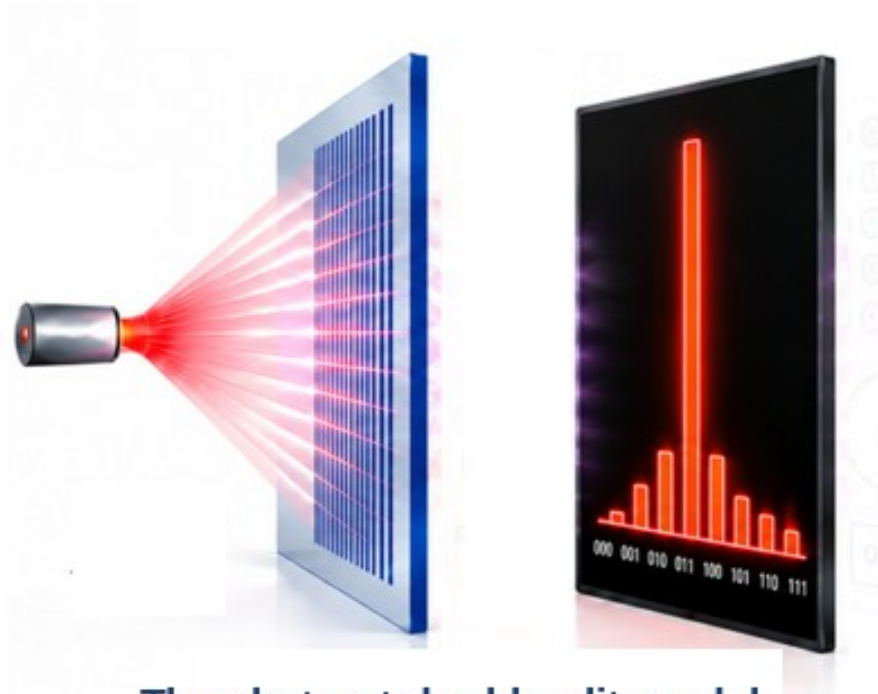


Q-LAB

Interdisciplinary Foundations for Reading and Running Quantum Algorithms

Boundary Objects · Interactional Expertise · Learning Scaffolds

A concise guide for physicists and chemists



The abstract double-slit model

Purpose

To help a domain researcher see one complete quantum workflow, preserve meaning across disciplinary handoffs, and move from recognition to structural fluency and guided application.

Project 2 · From the Double-Slit Model to Quantum Computing
Concept and visual design: Sarah Ibrahim Alkhamis · July 2026

Guide Map

Part	What you will learn
I. Why the whole map matters	Why familiar topics can still feel disconnected across a quantum workflow.
II. From Project One to Project Two	How the abstract double-slit model is scientifically bounded, and how the Mach-Zehnder interferometer bridges it to a one-qubit circuit.
III. Interdisciplinary foundations	Boundary objects, boundary crossing, translation, core Translation Cards, scaffolds, and interactional expertise.
IV. Quantum education cases	What interdisciplinary teaching cases in quantum computing actually show.
V. The Q-Lab framework	How the eight stages, detailed lens question sets, canonical programmatic path, path alignment, transpilation, and quality rail work together.
VI. How to use the scaffold	How to move through the guide and QPE notebook at different levels of depth.
VII. Guided application: Minimal QPE	How one phase-estimation workflow moves from a declared object to evidence and meaning.
VIII. Illustrative cross-lens reading	How a published SWAP-less QFT case reveals cross-lens diagnosis, upstream intervention, and contemporary signals from the field.
IX. Conclusion and references	The central idea and sources for further study.

Core promise

This guide does not replace mathematics, programming, or domain expertise. It helps the reader see how they connect inside one complete quantum workflow.

Part I - Why the Whole Map Matters

Quantum computing is often encountered as a sequence of local objects: Hilbert spaces, Hamiltonians, gates, circuits, APIs, backends, counts, estimators, and plots. Each object can be familiar while the complete scientific workflow remains difficult to see.

For a physicist or chemist, the central problem is often continuity. The same scientific idea changes language and form at every handoff:

- A molecule becomes a Hamiltonian; the Hamiltonian becomes Pauli terms; the Pauli terms become software objects and circuits.
- A logical circuit becomes a backend-compatible artifact; execution returns bitstrings, counts, memory, and metadata.
- Raw records become reconstructed quantities; reconstructed quantities become bounded scientific claims.
- At every transition, meaning can be lost even when the code runs.

The structural question

How can a learner or interdisciplinary team read a quantum algorithm as one evidence-bearing scientific workflow rather than as an isolated circuit?

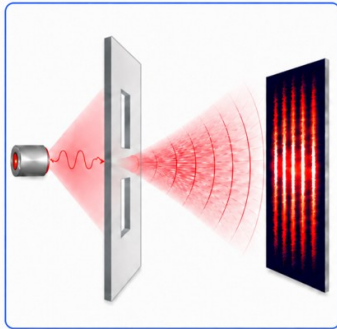
What this guide provides

A shared structure for locating the scientific question, the problem-carrying object, the executable artifacts, the handoffs, the evidence, and the checks. The aim is not to simplify the science until it becomes easy. The aim is to make the transitions inspectable.

Part II - From Project One to Project Two

Project One showed that a double-slit experiment and a one-qubit interferometric circuit can share an amplitude-level structure under an explicit encoding. It grounded coherent alternatives, relative phase, recombination, measurement, and entanglement without claiming that slits are literally qubits.

Superposition — branches of ONE system



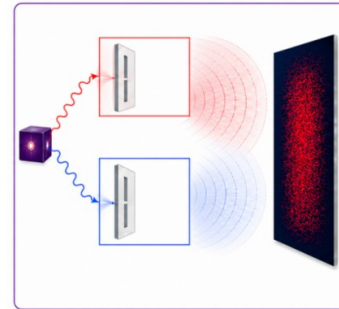
$$|\psi\rangle = \alpha |\psi_1\rangle + \beta e^{i\phi} |\psi_2\rangle$$

THE DOUBLE SLIT

Between the slits and the screen lies the playground: every single shot explores infinitely many alternatives, each contributing an amplitude with a well-defined RELATIVE phase. The two families interfere; one click lands per shot, weighted by the squared total — and repetition reveals the landscape.

Concept & visual design: Sarah Alkhamis

Entanglement — shared branches of a COMPOSITE system



$$|\Psi\rangle_{AB} = \alpha |\psi_1^A\rangle |\psi_1^B\rangle + \beta e^{i\phi} |\psi_2^A\rangle |\psi_2^B\rangle$$

THE SECOND-ORDER DOUBLE SLIT

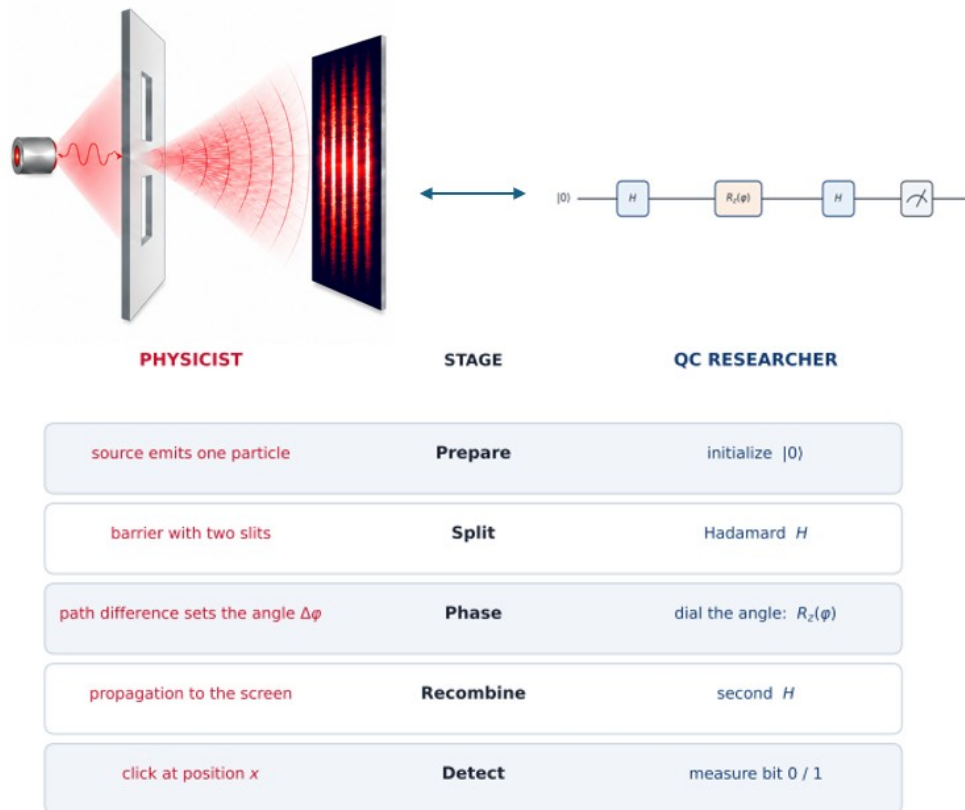
One conservation law selects two grand slits for the pair — and inside each, every system is still a first-order double slit of its own. For a maximally entangled pair, which-path detection on either partner acts on the grand slits — no local fringes; the phase lives in the joint correlations.

Concept & visual design: Sarah Alkhamis

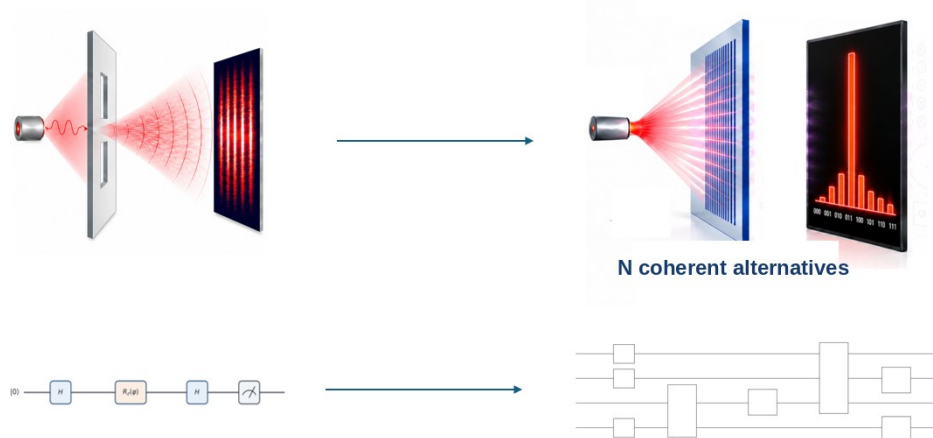
Project One recap: superposition as coherent branches of one system, and entanglement as shared branches of a composite system.

Project Two keeps the same discipline. The abstract double-slit model is the shared anchor, not a complete physical model of every quantum algorithm.

A Limited Translation



A limited translation between a two-path apparatus and a one-qubit interferometer.



Extension: N coherent alternatives; for n qubits, $N = 2^n$ basis states.

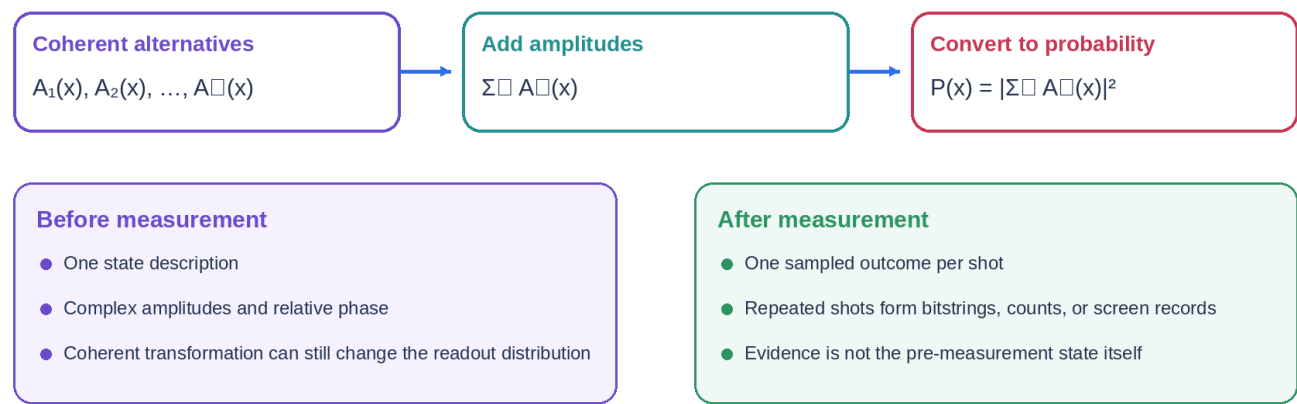
The abstraction contract

Keep coherent alternatives, complex amplitudes, relative phase, transformation before readout, and repeated probabilistic records. Leave behind literal slit geometry, a continuous screen, a specific particle, and the claim that every algorithm has one visible interference zone.

Scientific Contract of the Abstract Double-Slit Model

The model begins with two coherent alternatives and extends representationally to N coherent alternatives. For n qubits, $N = 2^n$ basis states. Its value lies in a bounded amplitude-level contract, not in a literal identification of slits with qubits.

Scientific contract of the abstract double-slit model



Visual Plate. Probabilities are obtained after coherent amplitudes are combined; repeated measurement records are evidence, not the pre-measurement state.

The model keeps	The model leaves behind
Coherent alternatives and complex amplitudes	Literal slit geometry and a continuous detection screen
Relative phase and phase-sensitive transformation	A specific particle, medium, or propagation law
Recombination before readout	The claim that every algorithm has one visible fringe pattern
One outcome per shot and a distribution across repetitions	The claim that the image is a complete physical model of every algorithm

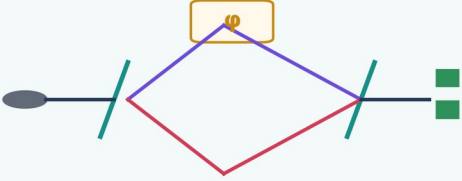
Scientific limit

The abstract double-slit model is an interferometric reading aid. It helps preserve the distinction between coherent state transformation and sampled evidence, while later mathematical and technical projects must supply the actual operators, encodings, approximations, and domain assumptions. (Feynman, Leighton, & Sands, 1965; Nielsen & Chuang, 2010)

Why the Abstract Double-Slit Model?

The Mach-Zehnder interferometer is a well-established two-mode model for split, relative phase, recombination, and detection, with a direct correspondence to a minimal one-qubit interferometric circuit. It is acknowledged here as a related model in the literature. This project instead selects the abstract double-slit model as its boundary object because it provides a wider and more flexible visual space for the full interdisciplinary workflow.

Related model in the literature
Mach-Zehnder interferometer



Precise two-mode interferometer
Clear correspondence with a minimal one-qubit circuit
split → phase → recombine → detect

Boundary object selected for this project
The abstract double-slit model



Starts with two coherent alternatives
Extends to N coherent alternatives and $N = 2^n$ basis states
Supports stages, lenses, artifacts, evidence, verification, and meaning

A deliberate representational choice — not a claim of physical equivalence or scientific superiority.

Visual Plate. The Mach-Zehnder interferometer is acknowledged as a related two-mode model; the abstract double-slit model is selected for its wider representational flexibility.

Feature	Mach-Zehnder interferometer	The abstract double-slit model
Primary role	A precise two-mode interferometric model.	The boundary object selected for this project.
Starting structure	Two explicit interferometric modes.	Two coherent alternatives.
Extension	Direct correspondence with a minimal one-qubit interferometric circuit.	Extends to N coherent alternatives and $N = 2^n$ basis states for n qubits.
What it visualizes best	Split, relative phase, recombination, and detection.	The wider journey: stages, lenses, artifacts, evidence, verification, and meaning.
Claim boundary	A related model acknowledged in the literature.	Selected for representational flexibility, not physical superiority.

Design choice
The abstract double-slit model was not selected because it is a more exact physical analogue of a quantum circuit. It was selected because it can organize a broader interdisciplinary reading while preserving the minimal interferometric structure.

Part III - The Interdisciplinary Foundations

1. Working meanings

Term	Working meaning in this guide
Disciplinary	A problem is framed and judged mainly through one field's objects, methods, and standards.
Multidisciplinary	Several disciplines contribute side by side; their outputs may be combined without deep integration.
Interdisciplinary	Representations, methods, or standards interact and change the way the problem is framed or solved.
Transdisciplinary	The work extends beyond academic disciplines to a shared practice around an applied or societal problem.

Q-Lab is primarily interdisciplinary: it makes the transformations between forms of reasoning visible while preserving the need for discipline-specific depth.

2. Boundary object

A boundary object is recognizable across different communities while remaining flexible enough to support their local needs and interpretations. It enables coordination without requiring complete agreement or identical vocabulary (Star & Griesemer, 1989; Star, 2010).

In this project, the boundary object is the abstract double-slit model. Its initial form contains two coherent alternatives. Its representational extension contains N coherent alternatives; for n qubits, $N = 2^n$ basis states. The model provides a shared anchor for discussing phase relations, transformation, readout, and evidence across disciplinary perspectives.

Important distinction

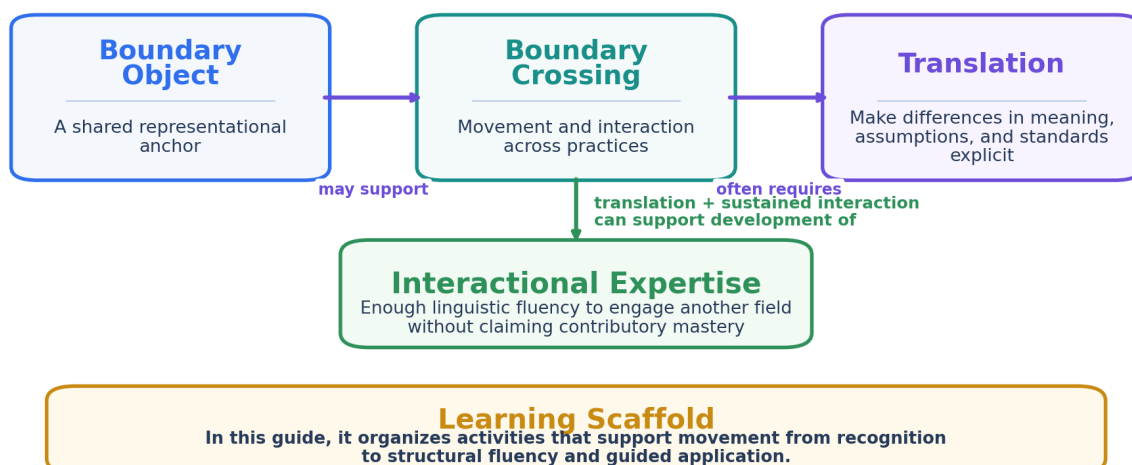
The boundary object is not the algorithmic journey itself. It is the shared object through which the journey becomes visible and discussable.

3. Boundary crossing

Boundary crossing refers to movement and interaction across different disciplinary or professional practices, especially when their objects, languages, methods, and standards do not fully align (Akkerman & Bakker, 2011). In a quantum workflow, a Hamiltonian may become a qubit operator; a logical circuit may become a compiled artifact; execution records may become a reconstructed estimate; and that estimate may become a bounded scientific claim.

How This Guide Brings the Foundations Together

Distinct research foundations, combined here as a project design synthesis



The concepts come from distinct research traditions. The arrangement above is proposed by this guide; it is not a single causal model established by the literature.

Translation across knowledge boundaries

A shared term can appear familiar while carrying different assumptions. Translation therefore involves more than replacing words: it makes objects, conventions, success criteria, and standards of evidence explicit. Carlile describes knowledge movement across boundaries through transfer, translation, and transformation (Carlile, 2004).

In Q-Lab, translation cards connect software terms to mathematical, physical, and evidential meanings. For example, phase can be read as a relative angle in an amplitude description, an information-bearing quantity in an algorithm, a gate parameter in code, and a quantity whose effect becomes visible only after recombination and measurement.

Practical boundary diagnosis

Before a handoff, ask: What changed representation? What information must be preserved? Which convention or assumption could be hidden? What check is sufficient before moving on?

Scaffolding and Interactional Expertise

4. Learning scaffold

A scaffold supports performance on a task that is currently beyond the learner's independent reach. It organizes attention, reduces avoidable load, exposes useful structure, and gradually transfers responsibility to the learner (Wood, Bruner, & Ross, 1976).

Here, the scaffold does not remove quantum complexity. It helps the learner locate that complexity, inspect the relevant interface, and decide which mechanism must be opened next.

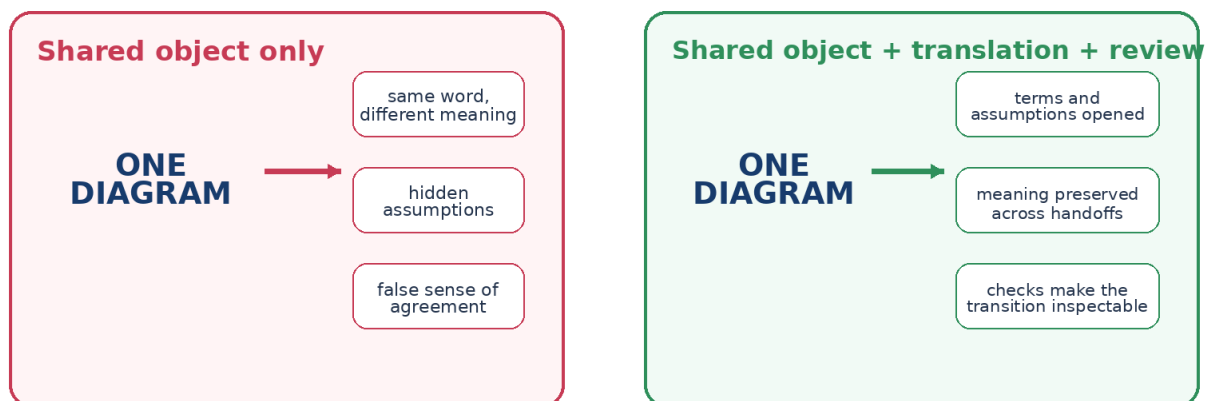
5. Interactional expertise

Interactional expertise is enough immersion in a field's language and reasoning to understand expert discourse and coordinate useful work without claiming contributory expertise in all of its practices (Collins & Evans, 2002; Collins, 2004).

Depth within interactional expertise	Observable capability
Recognition	Identify the active stage, lens, artifact, and missing piece.
Structural fluency	Explain how a decision in one stage constrains later stages and how meaning changes at interfaces.
Guided application	Inspect, run, reconstruct, and verify a supported workflow while knowing where specialist review is still required.

In Q-Lab, Recognition, Structural Fluency, and Guided Application are three degrees of depth inside interactional expertise. They are not separate kinds of expertise and do not imply contributory mastery.

A shared object is central to this scaffold - but not sufficient



Read more

Boundary objects: Star & Griesemer (1989); Star (2010). Boundary crossing: Akkerman & Bakker (2011). Knowledge boundaries: Carlile (2004). Interactional expertise: Collins & Evans (2002); Collins (2004). Scaffolding: Wood, Bruner, & Ross (1976).

Core Translation Cards

Translation Cards build interactional expertise by showing how the same term changes dialect while the underlying contract is preserved. They are not definitions that erase disciplinary differences; each card ends with an operational question that can be checked in the workflow.

Term	Physical reading	Quantum-computational reading	Operational question
Branch	One coherent alternative available to the system: a path, mode, level, or channel.	A basis component $ x\rangle$ of the register state.	Which alternatives are coherent, and what labels them?
Amplitude	A complex contribution with magnitude and phase.	Coefficient c_x in $ \psi\rangle = \sum_x c_x x\rangle$; $ c_x ^2$ becomes a measurement probability in the chosen basis.	Are amplitudes being combined before probabilities are formed?
Phase	A relative angle whose physical effect appears through interference.	A relative phase encoded by a unitary, rotation, oracle, or controlled evolution.	Where is phase introduced, and how is its effect made readable?
Measurement	A registered event or sampled value from an apparatus.	Sampling a bitstring or observable outcome from a specified basis.	What is measured, with how many shots, and what raw records are kept?
Entanglement	Correlations belonging to one composite state, not separate local mixtures.	A joint state that cannot be factored into independent subsystem states.	Which correlations are genuinely nonseparable, and how are they verified?

Translation rule

Translate meaning, assumptions, and evidence conditions - not vocabulary alone. A familiar word can conceal different objects, standards of correctness, or units of analysis.

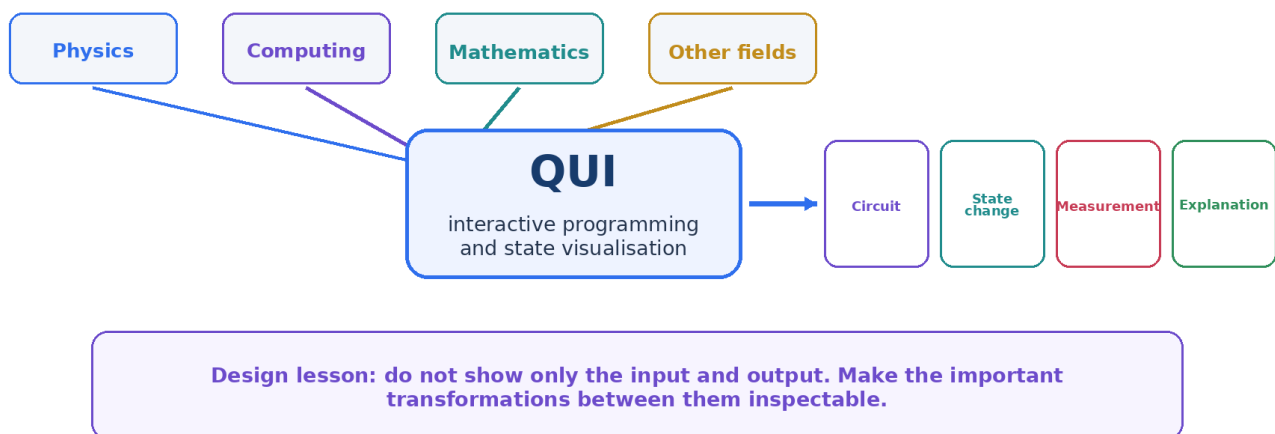
Part IV - Interdisciplinary Approaches in Quantum Computing Education

University of Melbourne: Elements of Quantum Computing and QUI

Elements of Quantum Computing is jointly offered by the University of Melbourne's School of Physics, School of Computing and Information Systems, and Centre for Brain, Mind and Markets. It introduces quantum computing to students from a range of backgrounds through quantum principles, programming, simulation, and access to online quantum systems (University of Melbourne, n.d.-a).

The university's Quantum User Interface (QUI) is integrated into Elements of Quantum Computing and Introduction to Quantum Computing. The subjects are open to students from all disciplines and assume no prior quantum-physics knowledge. QUI lets learners program a circuit and inspect the quantum state as the program develops rather than receiving only the final output (University of Melbourne, n.d.-b; Trounson, 2018).

University of Melbourne: one interdisciplinary entry point



The black-box concern - only in this case

The University of Melbourne describes the quantum computer as a black box that is difficult to inspect while it runs. QUI addresses that teaching problem by visualizing the state at successive stages of a program. The lesson for this guide is limited and practical: do not show only input and output; make the important transformations between them inspectable.

The public materials describe the design and classroom use of QUI. They should be treated as a teaching case, not as a controlled proof that visualization alone improves learning.

What Other Quantum-Education Cases Show

Case	What it contributes	Source
Interdisciplinary QIS classroom survey	Courses bring together varied instructor and student backgrounds. Common issues include differences in preparation, difficulty with mathematical formalism - especially linear algebra - and the need to balance accessibility with rigor. No single canonical introductory course emerged.	Meyer et al. (2022)
Practical software-driven course	An undergraduate course centered quantum algorithms around hands-on programming, self-paced exercises, assignments, and a final project. Students reported that programming exercises and the project contributed strongly to their learning.	Mykhailova & Svore (2020)
Quantum Interactive Learning Tutorial	A guided inquiry tutorial was developed and validated around common student difficulties, then used in two courses. The study reported improved performance on the concepts addressed by the tutorial.	Hu, Li, & Singh (2024)

What these cases support

- Multiple entry routes are necessary because learners do not begin with the same disciplinary preparation.
- Hands-on programming and shared artifacts can connect formal ideas to executable actions.
- Visual and inquiry-based support should expose structure without replacing mathematical depth.
- An interdisciplinary course still needs explicit choices about prerequisites, representations, evidence, and rigor.

What they do not prove

These cases support the diagnosis behind Q-Lab, but they do not prove that this particular visual scaffold is effective. Q-Lab remains a human-designed educational project and practical companion to the notebook.

Part V - The Q-Lab Framework



The shared algorithmic body and the three supporting structures used to read it.

The boundary object is the abstract double-slit model. Its initial form is two coherent alternatives; its representational extension is N coherent alternatives; for n qubits, $N = 2^n$ basis states. The eight stages are the shared conceptual body. The five lenses show which forms of reasoning are active. The programmatic path realizes the journey through executable artifacts. The quality rail asks what could fail and what evidence must be preserved at every transition.

Key distinction

The boundary object, the stages, the lenses, the code path, and the quality rail are different tools. They should not be collapsed into one diagram or treated as interchangeable.

The Eight Shared Coordinates

1. Origin	Question: What is the domain question, target quantity, scope, and success criterion? Output: Problem statement and acceptance rule.
2. Representation	Question: What mathematical object carries the problem? Output: Basis, state space, Hamiltonian, unitary, oracle, graph, matrix, channel, constraints, and encoding.
3. Transformation	Question: What process changes the state or imprints the problem information? Output: Unitary, oracle call, ansatz layer, controlled evolution, or logical protocol.
4. Interference	Question: How do coherent amplitudes and relative phases shape the readable result? Output: Phase-sensitive redistribution, selection, or basis change.
5. Measurement	Question: What is measured, in which basis, and with what sampling design? Output: Readout and observable plan.
6. Evidence	Question: What raw records and provenance return from execution? Output: Counts, bitstrings, samples, job data, shots, versions, layout, and metadata.
7. Verification	Question: Given the reconstructed quantity, how is it compared with a reference, tested, and assigned uncertainty? Output: Baseline comparison, diagnostics, uncertainty, error bars, and an acceptance decision.
8. Meaning	Question: What can responsibly be said in the language of the original problem? Output: Bounded domain statement with assumptions, uncertainty, and limitations.

The Five Analytical Lenses



Lens	Primary questions
Domain & Physical	What phenomenon or scientific task is represented? What quantity matters? Is the result physically plausible?
Mathematical Modelling	What object carries the problem? What structure, symmetry, basis, approximation, and precision matter?
Quantum Computation	How are states prepared, information imprinted, amplitudes transformed, ancillas managed, and readout designed?
Classical Computation & Evidence	What decoding, estimator, baseline, statistics, uncertainty, interface checks, and post-processing are required?
Execution & Hardware	What changes under decomposition, routing, native gates, noise, shots, runtime, mitigation, and error correction?

The correct question

Not: Who owns this stage? Instead: Which forms of reasoning and checking are active here, and who is qualified to lead and review them in this case?

Reading the Lenses as Structured Question Sets

Each lens contains topics, and each topic generates a characteristic question. The following matrices show where those questions become active across the shared conceptual body and the canonical programmatic path. Their purpose is not to prove orthogonality or assign permanent ownership. Their purpose is to make disciplinary attention explicit so that later interaction can be designed rather than improvised.

Lens	Central concern	Boundary that keeps it distinct
Domain & Physical	Phenomenon, target quantity, physical plausibility, and return to the original problem.	Do not replace physical meaning with a derivation or circuit diagram.
Mathematical Modelling	Problem-carrying objects, structure, symmetry, encoding, approximation, and precision.	Do not turn formal structure into implementation detail.
Quantum Computation	How states, amplitudes, phases, ancillas, and readout are engineered coherently.	Do not collapse quantum logic into general physical description or backend operations.
Classical Computation & Evidence	Decoding, estimators, baselines, statistics, reproducibility, interfaces, and conditional optimization.	Do not make it a residual box for everything non-quantum.
Execution & Hardware	Backend adaptation, connectivity, native instructions, noise, runtime, mitigation, and error correction.	Do not redesign the logical algorithm while pretending to only execute it.

How to use the tables

Read each row as: What topic is active? What question does that topic ask here? Which other lens must lead or review the handoff? Blank cells are informative: a lens may be conditional or genuinely inactive for a particular algorithm.

Lens 1 - Domain & Physical

Central question: What phenomenon or scientific task is represented, what quantity matters, and what can the final value mean in the original domain?

Boundary rule: Keep the reading descriptive and domain-grounded. The physical lens may refer to equations supplied by the model, but it should not become the mathematical derivation, circuit design, or backend implementation.

Topics across the shared conceptual body

Topic	Orig.	Rep.	Trans.	Interf.	Meas.	Evid.	Verif.	Meaning
Target phenomenon	•							
Physical meaning of the representation		•						
Where problem information enters physically			•	•				
Physical quantity being measured					•			
Physical plausibility of the result							•	
Return to the phenomenon								•

A mark means the topic is genuinely active. Blank cells should remain blank rather than being filled by force.

Topics across the canonical programmatic path

Topic	Build	Translate	Validate	Transpile	Execute	Preserve Evidence	Reconstruct
Target phenomenon	•						
Physical meaning of the representation	•						
Where problem information enters physically	•						
Physical quantity being measured	•				•		
Physical plausibility of the result							
Return to the phenomenon							

How to read this lens

- The lens begins at Origin and closes at Meaning. It preserves the scientific thread before and after the technical middle of the workflow.
- At Interference it asks where phase or coherent alternatives arise in the phenomenon; the Quantum Computation lens asks how that structure is engineered in the circuit.
- Physical plausibility and domain meaning occur after Reconstruct, inside Verification and Meaning; they are intentionally not assigned to Reconstruct.

Lens 2 - Mathematical Modelling

Central question: What mathematical object carries the problem, and which structure, basis, approximation, and precision make the task well defined?

Boundary rule: Stay with formal objects, mappings, invariants, and error conditions. Do not drift into gate-by-gate circuit engineering or hardware-specific compilation.

Topics across the shared conceptual body

Topic	Orig.	Rep.	Trans.	Interf.	Meas.	Evid.	Verif.	Meaning
Problem type and target	•							
Problem-carrying object		•						
Structure, symmetry, and constraints		•						
Encoding and basis choice		•	•					
Approximation and precision	•	•					•	
Success criterion	•						•	

A mark means the topic is genuinely active. Blank cells should remain blank rather than being filled by force.

Topics across the canonical programmatic path

Topic	Build	Translate	Validate	Transpile	Execute	Preserve Evidence	Reconstruct
Problem type and target	•						
Problem-carrying object	•						
Structure, symmetry, and constraints	•		•				
Encoding and basis choice	•		•				
Approximation and precision	•						•
Success criterion	•						

How to read this lens

- The lens concentrates around Origin and Representation because this is where the problem is made mathematically explicit.
- Encoding is a boundary topic: Mathematical Modelling specifies what the variables and basis mean; Quantum Computation determines how that encoding is realized coherently.
- Reconstruct may calculate a phase-bin width or decoded numerical quantity. Whether the declared success criterion is met remains a Verification decision.

Lens 3 - Quantum Computation

Central question: How are states, amplitudes, phases, entanglement, and measurement bases engineered so that problem structure becomes readable?

Boundary rule: Focus on coherent logical design. Do not replace the domain meaning, abstract mathematical specification, or device-adaptation problem.

Topics across the shared conceptual body

Topic	Orig.	Rep.	Trans.	Interf.	Meas.	Evid.	Verif.	Meaning
State preparation		•	•					
Problem-information imprint			•					
Coherent transformation and interference			•	•				
Entanglement, ancillas, and reversibility			•	•				
Readout and measurement design					•			
Logical correctness and logical resources							•	

A mark means the topic is genuinely active. Blank cells should remain blank rather than being filled by force.

Topics across the canonical programmatic path

Topic	Build	Translate	Validate	Transpile	Execute	Preserve Evidence	Reconstruct
State preparation	•						
Problem-information imprint	•						
Coherent transformation and interference	•						
Entanglement, ancillas, and reversibility	•						
Readout and measurement design	•				○		
Logical correctness and logical resources	•		•				

How to read this lens

- This lens concentrates in Transformation and Interference, and most of its programmatic work is contained inside Build - specifically Build Circuit.
- The physical lens asks where phase appears in the phenomenon; the quantum-computational lens asks how gates and controlled operations make phase readable.
- Measurement has two layers: Quantum Computation designs the informative basis and readout logic; Execution & Hardware determines shots, backend behavior, and runtime execution.

Lens 4 - Classical Computation & Evidence

Central question: How are raw records decoded, compared, quantified, reproduced, and - when required - used inside a classical-quantum loop?

Boundary rule: Do not treat this lens as a residual category. Separate what belongs to Reconstruct, what belongs to Verification, what crosses the whole quality rail, and what appears only in particular algorithm families.

Topics across the shared conceptual body

Topic	Orig.	Rep.	Trans.	Interf.	Meas.	Evid.	Verif.	Meaning
Classical baseline							•	
Decoding and estimator						•	•	
Post-processing and statistics						•	•	
Uncertainty and confidence							•	
Testing and reproducibility	•	•	•	•	•	•	•	•
Decomposition and interface contracts	•	•	•	•	•	•	•	•
Optimizer - conditional			•		•		•	

A mark means the topic is genuinely active. Blank cells should remain blank rather than being filled by force.

Topics across the canonical programmatic path

Topic	Build	Translate	Validate	Transpile	Execute	Preserve Evidence	Reconstruct
Classical baseline							
Decoding and estimator							•
Post-processing and statistics							•
Uncertainty and confidence							○
Testing and reproducibility	•	•	•	•	•	•	•
Decomposition and interface contracts	•	•	•	•	•	•	•
Optimizer - conditional	↻				↻		↻

How to read this lens

- Decoding, estimators, post-processing, and descriptive statistics belong mainly to Reconstruct. Classical baselines, uncertainty judgments, and acceptance belong to Verification after the canonical path.
- Testing, reproducibility, and interface contracts cross every transition; they are qualities of the workflow rather than a single stage.
- The optimizer is conditional. In VQE, QAOA, or variational time evolution it forms a loop between Execute, Reconstruct, and parameter updates; it should not be forced into QPE or every other algorithm.

Lens 5 - Execution & Hardware

Central question: How must the logical circuit change to execute on a target system, and how do connectivity, native operations, noise, runtime, and error control constrain the evidence?

Boundary rule: Handle backend adaptation and execution without silently redefining the scientific task or logical algorithm. Any such redesign must be surfaced as a cross-lens intervention.

Topics across the shared conceptual body

Topic	Orig.	Rep.	Trans.	Interf.	Meas.	Evid.	Verif.	Meaning
Backend constraints and basis gates			•	•				
Layout, connectivity, routing, and depth			•	•				
Noise modelling and mitigation					•	•	•	
Error-correction and fault-tolerance awareness					•		•	•
Shots, runtime, and monitoring					•	•		
Calibration, job context, and provenance						•	•	

A mark means the topic is genuinely active. Blank cells should remain blank rather than being filled by force.

Topics across the canonical programmatic path

Topic	Build	Translate	Validate	Transpile	Execute	Preserve Evidence	Reconstruct
Backend constraints and basis gates				•	◦		
Layout, connectivity, routing, and depth				•			
Noise modelling and mitigation				•	•		◦
Error-correction and fault-tolerance awareness				•	•		
Shots, runtime, and monitoring					•	•	
Calibration, job context, and provenance					•	•	◦

How to read this lens

- The lens concentrates in Transpile and Execute, then remains active in Preserve Evidence because calibration, layout, backend, shots, and runtime are part of provenance.
- Quantum Computation designs the measurement; Execution & Hardware realizes it under a particular backend and sampling budget.
- Hardware limits continue to bound Meaning: a scientifically plausible value must still be stated with the noise, mitigation, and error-correction conditions that produced it.

Lens Fingerprints and High-Value Interaction Points

The matrices are question maps. They show where a field's topics become active and where another field must join the reasoning. The patterns are not a proof of orthogonality, and the cells are not an ownership chart.

Lens	Conceptual centre of gravity	Programmatic centre of gravity
Domain & Physical	Origin and Meaning, with checks at Measurement and Verification	Build and Execute; scientific review resumes after Reconstruct
Mathematical Modelling	Origin and Representation	Build, with structural checks in Validate and numerical output in Reconstruct
Quantum Computation	Transformation, Interference, and measurement design	Build - especially Build Circuit - plus logical checks in Validate
Classical Computation & Evidence	Evidence-to-Verification interface and the quality rail	Reconstruct, cross-cutting tests and interfaces; optional variational loop
Execution & Hardware	Implementation around coherent processing, Measurement, Evidence, and Verification	Transpile, Execute, and Preserve Evidence

Where interaction becomes especially valuable

Shared location	Active lenses	Questions that must be kept distinct and then connected
Representation	Domain & Physical + Mathematical Modelling + Quantum Computation	What do states mean? What object and basis carry the problem? How will that encoding be prepared and manipulated?
Interference	Domain & Physical + Quantum Computation	Where do coherent alternatives or phase relations arise physically? How is their recombination engineered logically?
Measurement	Domain & Physical + Quantum Computation + Execution & Hardware	What quantity matters? Which basis is informative? How many shots and which backend execute the readout?
Evidence and Reconstruct	Classical Computation & Evidence + Execution & Hardware	Which raw records and provenance are retained? How are they decoded without yet claiming verification?
Verification	All five lenses	Is the quantity plausible, mathematically within tolerance, logically correct, statistically supported, and consistent with hardware evidence?
Meaning	Domain & Physical + Classical Computation & Evidence + Execution & Hardware	What bounded scientific claim follows, and which uncertainty and device conditions limit it?

Why this matters for creativity

A cross-lens idea becomes visible when a problem diagnosed under one lens can be traced to an earlier decision governed by another. The tables prepare that move by making the questions and interfaces explicit.

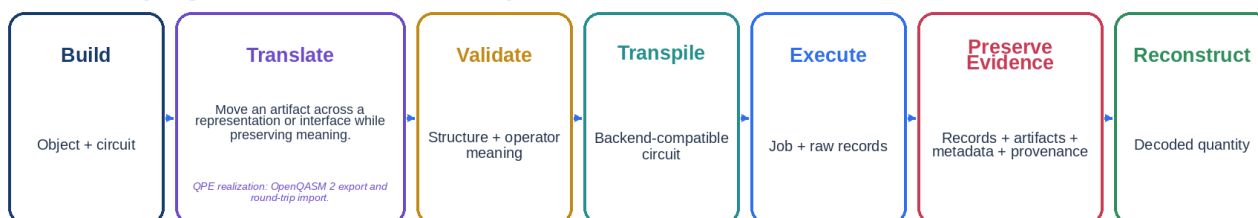
The Programmatic Path and Quality Rail

Two connected paths

Shared conceptual body

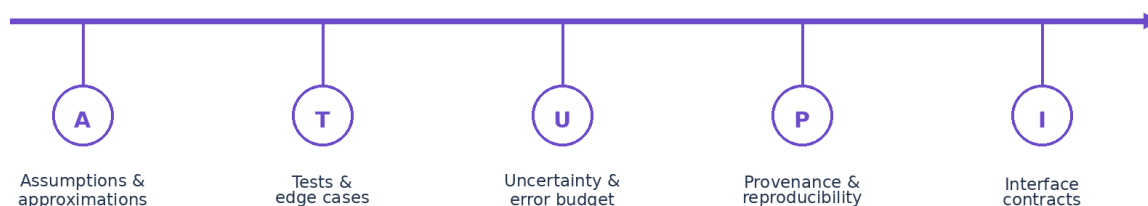


Canonical programmatic and execution path



Translate moves an artifact across a representation or interface while preserving meaning; in the QPE notebook, this is realized through an OpenQASM 2 export and round-trip import. Validate asks whether an artifact is structurally and semantically well formed. Reconstruct converts retained records into a decoded quantity. Verification then tests that quantity against a reference, assigns uncertainty, and asks whether the evidence supports the bounded scientific claim. Reconstruct does not subsume verification or meaning.

Cross-cutting quality rail



At every transition ask

What is the claim? What could fail? What check would reveal it? What evidence must be retained? What threshold allows the workflow to move on?

The Two Paths Are Aligned - Not Identical

The shared conceptual body describes scientific coordinates. The canonical programmatic path describes executable artifacts. Their relationship is many-to-many: one programmatic step can realize several conceptual coordinates, and one conceptual coordinate can require several artifacts and checks.

Canonical step	Internal artifact or action	Main conceptual coordinates	Central question
Build	Build Object + Build Circuit	Origin, Representation, Transformation, Interference, and measurement design	Which problem object and logical circuit realize the declared mission?
Translate	Move an artifact across a representation or interface while preserving meaning	Boundary between logical design, software representations, and execution interfaces	What must remain invariant when the artifact changes form?
Validate	Structural, semantic, and operator-level checks	Quality checks across Representation and Transformation	Is the artifact well formed, and does it preserve the intended operation?
Transpile	Backend-compatible circuit	Device realization of Transformation and Interference before Measurement	How does the logical circuit fit the target instruction set and topology?
Execute	Backend job and raw sampled records	Measurement	What was run, under which settings, and what raw records returned?
Preserve Evidence	Counts, bitstrings, memory, artifacts, metadata, versions, layout, and provenance	Evidence	What must be retained so the run remains inspectable and reproducible?
Reconstruct	Decoded quantity and resolution information	Crossing from Evidence toward Verification	How are retained records converted into a quantity that can later be tested?

Strict separation

Reconstruct converts records into a decoded quantity. Verification compares that quantity with references, assigns uncertainty, tests evidence consistency, and applies the acceptance rule. Meaning then returns the verified result to the original scientific question.

Inside Transpile: A Qiskit-Style Six-Stage Pipeline

Transpile is the canonical step that adapts a logical circuit to a target backend. In Qiskit, the preset staged pass manager is organized into six named stages. Other toolchains may package the work differently, so this page is a technical realization rather than a universal decomposition.

Two meanings of translation
Canonical Translate means moving any artifact across a representation or interface while preserving meaning. Qiskit's translation stage is narrower: it is a substage inside Transpile that converts gates to the target basis instructions.

Qiskit stage	What it does	Key question	Evidence or condition to retain
1. init	Unroll custom or multi-qubit instructions and perform initial logical lowering, usually to one- and two-qubit operations.	Can every high-level operation be represented by supported lower-level primitives?	Input circuit, decomposition choices, pass-manager configuration
2. layout	Map logical qubits to physical qubits on the target.	Which physical qubit should carry each logical qubit?	Initial layout, target calibration context, seed or heuristic settings
3. routing	Insert operations such as SWAPs when required by limited connectivity.	How are non-neighbour interactions made executable?	Final layout/permutation, inserted SWAPs, routing method, depth increase
4. translation	Convert operations to the backend instruction set or basis gates.	Are all operations expressed in the target's native language?	Target basis/ISA, synthesis method, operator-equivalence checks
5. optimization	Combine, cancel, or resynthesize operations to reduce cost while preserving the circuit's action.	Can depth or gate count be reduced without changing meaning?	Optimization level, cost metrics, before/after resources, random seed
6. scheduling	Make wall-clock timing and idle periods explicit when scheduling is requested.	When does each operation occur, and how are timing constraints handled?	Scheduling method, instruction durations, delays, timing constraints

Source: IBM Quantum Documentation, "Introduction to transpilation" and "Transpiler stages" (accessed July 2026).

Why routing matters to cross-lens reasoning
A large SWAP overhead appears under the Execution & Hardware lens, but the strongest intervention may occur earlier - in encoding, representation, or circuit synthesis. That is the pattern illustrated by the SWAP-less QFT case.

Part VI - How to Use the Scaffold

Mode	What the learner does
Quick Run	Follow the programmatic path and execute the code cells in order.
Deep Read	Use the shared stages, active lenses, and Translation Cards to understand what each artifact means.
Research Mode	Inspect validation, evidence, provenance, reconstruction, and quality checks before accepting the final statement.

A practical reading routine

- Start from the mission: state the quantity to be inferred, the declared model, and the acceptance criterion.
- Map the shared conceptual body: Origin, Representation, Transformation, Interference, Measurement, Evidence, Verification, and Meaning.
- Trace the canonical programmatic path: Build -> Translate -> Validate -> Transpile -> Execute -> Preserve Evidence -> Reconstruct.
- Use the active lenses: identify which lens is primary, supporting, or handoff-aware at each stage.
- Read the Translation Card before the code: connect the software term to its mathematical, physical, or evidential meaning.
- Apply the quality question: ask what could fail, what must remain unchanged, and what evidence must be retained.
- Separate raw evidence, reconstructed estimate, verified result, and bounded meaning.

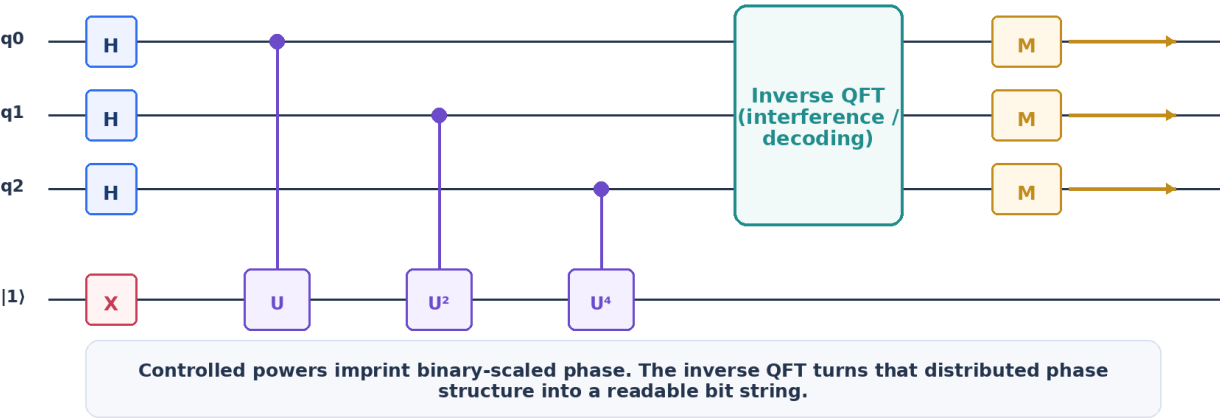
A good scaffold output

The learner can identify the current conceptual stage, programmatic artifact, active lens, interface contract, retained evidence, quality check, and limit of the final statement.

Part VII - Guided Application: Minimal Quantum Phase Estimation

The notebook estimates the exactly representable phase $\phi = 3/8 = 0.011_2$ using three evaluation qubits, the eigenstate $|1\rangle$, and the unitary $U = P(2\pi\phi)$. The setup uses Qiskit 2.5.1, 2048 shots, fixed seeds, and Qiskit's built-in BasicSimulator.

Minimal Quantum Phase Estimation: coherent logic



Shared stage	QPE reading
Origin	Infer an eigenphase ϕ to three-bit precision under a declared ideal model.
Representation	Use $U \psi\rangle = \exp(2\pi i\phi) \psi\rangle$, the eigenstate $ 1\rangle$, and three evaluation qubits.
Transformation	Controlled powers U , U^2 , and U^4 accumulate phase at binary scales.
Interference	The inverse QFT converts distributed phase structure into amplitudes concentrated at the binary estimate.
Measurement	Measure the three-qubit evaluation register.
Evidence	Preserve counts, shot memory, OpenQASM 2 artifacts, backend settings, versions, seeds, compiled metrics, and provenance.
Verification	Given the reconstructed phase estimate, compare it with the exact reference, calculate wrapped error and success probability, and apply the acceptance rule.
Meaning	Interpret $\hat{\phi} = 3/8$ as the eigenphase of the declared unitary; any physical interpretation requires an explicit model.

QPE as an Artifact and Evidence Chain

Canonical step	QPE artifact
Build	Problem object and logical circuit.
Translate	OpenQASM 2 export and re-imported round-trip circuit.
Validate	Structural checks and measurement-free operator equivalence up to global phase.
Transpile	Backend-compatible circuit.
Execute	Job, counts, and shot memory.
Preserve Evidence	JSON evidence bundle and retained artifacts.
Reconstruct	Dominant bitstring, decoded phase estimate, and phase-bin width.

Notebook substeps: Build contains Build Object and Build Circuit. Translate contains the OpenQASM 2 export and the re-imported round-trip circuit. Preserve Evidence saves counts, memory, artifacts, metadata, and provenance.

Selected Translation Cards

Software term	Software meaning	Scientific meaning
PhaseGate	A parameterized one-qubit unitary	The unitary whose eigenstate $ 1\rangle$ acquires the declared eigenphase.
Controlled U^{2k}	Conditional powers of the unitary	Coherent phase accumulation at binary scales.
Counts / memory	Aggregated frequencies / every shot record	Raw measurement evidence before reconstruction.
Wrapped error	Circular difference on the interval $[0,1)$	A phase-aware verification metric.

Quality checkpoints

Transition	Main check
Build	Do the problem object, phase convention, register roles, and controlled powers match the declared mission?
Translate / Validate	Did the circuit survive the OpenQASM 2 round trip structurally and preserve the measurement-free operator up to global phase?
Transpile / Execute	Which backend, compiled artifact, shots, and seeds generated the records?
Preserve Evidence	Can every record and artifact be traced back to the mission and execution context?
Reconstruct	Does bit order and binary decoding match the circuit convention?
Verification	Do the wrapped error, success probability, reference comparison, and evidence consistency satisfy the declared acceptance rule?

Verification, Meaning, and the Limit of the Example

Verification

Under the declared bit-order convention, verification passes when the dominant bitstring is 011, the reconstructed phase is $\hat{\phi} = 3/8$, the wrapped error is zero within numerical tolerance, the success probability satisfies the declared threshold, and the preserved evidence chain remains consistent.

Meaning

The decoded value is an eigenphase of the declared unitary. It becomes an energy, frequency, or another physical quantity only when an explicit model and conversion relation are supplied.

What the evidence supports

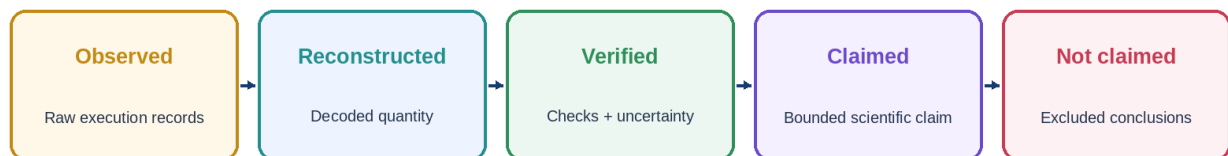
Supported statement

For the declared unitary $U = P(2\pi\phi)$, prepared eigenstate $|1\rangle$, ideal BasicSimulator, three evaluation qubits, 2048 shots, and preserved settings, the workflow reconstructs $\hat{\phi} = 3/8$.

What the evidence does not support

- Quantum advantage.
- Real-hardware accuracy.
- Estimation of an unknown physical energy.
- Correctness for arbitrary unitaries or arbitrary input states.

A physical use of $U = \exp(-iHt)$ would additionally require a Hamiltonian model, an eigenstate-preparation strategy, explicit sign and time conventions, and a broader uncertainty analysis.

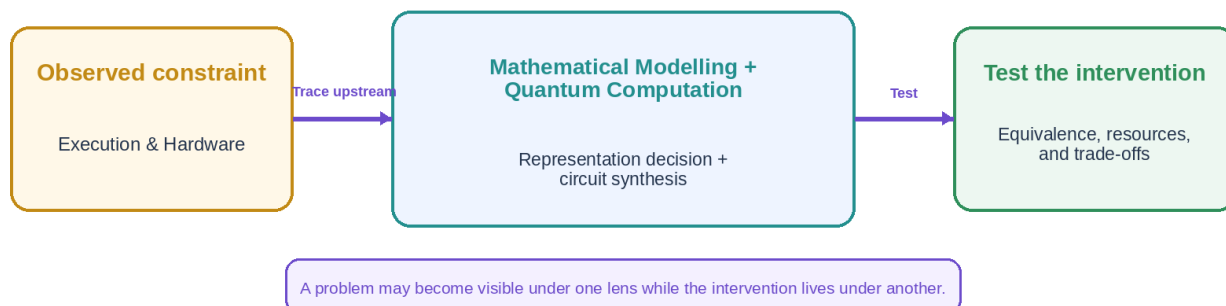


Part VIII - Illustrative Cross-Lens Reading

The scaffold can be used operationally to run a complete workflow, as in the QPE notebook. It can also be used conceptually to re-read a published result through the lenses. Cross-lens reasoning is not a sixth lens: it is the movement of a diagnosed problem toward an earlier decision governed by another lens.

In the SWAP-less QFT case, limited hardware connectivity is the observed constraint under the Execution & Hardware lens. The published intervention tracks parity information and changes an earlier representation decision and circuit-synthesis strategy, activating Mathematical Modelling and Quantum Computation. The framework did not produce this result; it makes the cross-lens movement inspectable (Klaver et al., 2026).

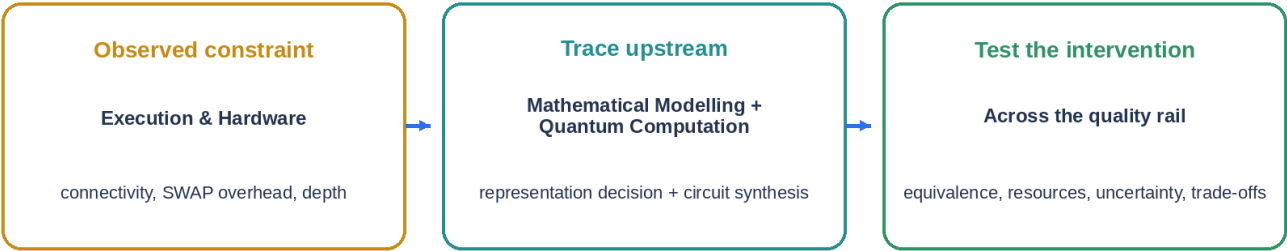
Illustrative cross-lens reading



The carousel provides the fuller visual reading of this case. Here, the example only establishes the second use of the scaffold: tracing a problem upstream across lenses and then testing equivalence, resources, and trade-offs.

Cross-Lens Reasoning as a Design Move

Cross-lens reasoning is a movement, not a sixth lens



The framework did not produce the published result. It makes the movement between diagnosis and intervention inspectable.

In the published SWAP-less implementation, limited connectivity and SWAP overhead are visible under Execution & Hardware. The intervention does not merely improve routing after the circuit is fixed. Parity-label tracking recodes how logical information is carried and synthesizes the QFT accordingly. The published linear-nearest-neighbour construction reports depth $5n - 3$ with $n^2 - 1$ CNOT gates. (Klaver et al., 2026)

Move	Question	SWAP-less QFT illustration
1. Diagnose	Where is the observable constraint and which metric makes it real?	Connectivity, SWAP count, CNOT count, and circuit depth under the Execution & Hardware lens.
2. Re-describe	How do the other lenses describe the same problem?	Mathematical Modelling asks how information is encoded; Quantum Computation asks how parities and coherent operations are synthesized.
3. Trace upstream	Which earlier decision creates the downstream cost?	The representation and synthesis strategy determine whether information must be physically swapped.
4. Relocate the intervention	What is the earliest decision that can change without breaking the scientific contract?	Change the encoding/label flow and circuit synthesis rather than treating SWAP insertion as inevitable.
5. Test	What must remain equivalent, and which new trade-offs appear?	Verify the implemented unitary, gate count, depth, connectivity assumptions, qubit overhead, and measurement decoding.
6. Bound the claim	What did the framework contribute?	The framework did not produce the published result; it makes the movement between diagnosis and intervention inspectable.

Creativity principle
Cross-lens interaction is not an extra cell in every matrix. It emerges when the lenses are placed on the same problem and a constraint can be moved to a different decision layer.

Contemporary Signals Through the Lenses

These cases are brief signals, not evidence that Q-Lab caused the results or that quantum advantage has been established. They show why the lenses remain useful as the field changes.

Contemporary case	What changed	Lenses made visible	Responsible reading
50-qubit QFT benchmark on IBM Heron r3 - preprint (2026)	A parity-architecture implementation was benchmarked at larger qubit count, making fidelity, instruction set, and architecture-algorithm co-design central.	Quantum Computation ↔ Execution & Hardware, with representation/synthesis decisions upstream.	Treat as a current preprint and hardware benchmark, not a general proof of practical advantage. (Aumann et al., 2026)
Hybrid quantum-classical KRAS inhibitor design (2025)	A 16-qubit quantum generative component was integrated with classical generative modelling; molecules were synthesized and experimentally evaluated.	Domain & Physical ↔ Quantum Computation ↔ Classical Computation & Evidence.	The value lies in the full evidence chain from model to experimental assays; comparisons do not establish broad quantum advantage. (Ghazi Vakili et al., 2025)
Toric-/surface-code error-correction progress (2025-2026)	Peer-reviewed work demonstrated below-threshold surface-code behaviour, while a 2026 preprint reported repeated toric-code syndrome extraction over many cycles.	Execution & Hardware ↔ Classical Computation & Evidence, with Meaning bounded by code distance, decoder, and noise model.	Separate peer-reviewed results from preprints and preserve the exact code, cycle count, decoder, and threshold definition. (Google Quantum AI, 2025; Atom Computing et al., 2026, preprint)

Use of current examples

The examples feed existing lenses; they do not become new stages. Their role is to show that representation, coherent design, classical evidence, and hardware constraints continue to interact in live research.

Conclusion

Q-Lab is a simple human-designed interdisciplinary scaffold for reading a quantum algorithm as one connected workflow. It brings together the abstract double-slit model as a shared boundary object, eight conceptual stages, five analytical lenses, a canonical programmatic path, and a cross-cutting quality rail.

The QPE notebook demonstrates the operational use of the framework. The detailed lens tables make each field's topics and questions explicit; the SWAP-less QFT reading shows how those lenses can interact when a constraint is traced upstream to a different design decision.

The goal

Help the learner keep the scientific thread while moving between physics, mathematics, quantum computation, software, hardware, and evidence.

Designed and developed by Sarah Ibrahim Alkhamis · July 2026

Interdisciplinary and Educational Foundations

1. Akkerman, S. F., & Bakker, A. (2011). Boundary crossing and boundary objects. *Review of Educational Research*, 81(2), 132-169.
2. Carlile, P. R. (2004). Transferring, translating, and transforming: An integrative framework for managing knowledge across boundaries. *Organization Science*, 15(5), 555-568.
3. Collins, H. M. (2004). Interactional expertise as a third kind of knowledge. *Phenomenology and the Cognitive Sciences*, 3, 125-143.
4. Collins, H. M., & Evans, R. (2002). The third wave of science studies: Studies of expertise and experience. *Social Studies of Science*, 32(2), 235-296.
5. Star, S. L. (2010). This is not a boundary object: Reflections on the origin of a concept. *Science, Technology, & Human Values*, 35(5), 601-617.
6. Star, S. L., & Griesemer, J. R. (1989). Institutional ecology, translations and boundary objects. *Social Studies of Science*, 19(3), 387-420.
7. Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, 17(2), 89-100.

Quantum Computing and Technical References

8. Cleve, R., Ekert, A., Macchiavello, C., & Mosca, M. (1998). Quantum algorithms revisited. *Proceedings of the Royal Society A*, 454, 339-354.
9. Hu, P., Li, Y., & Singh, C. (2024). Investigating and improving student understanding of the basics of quantum computing. *Physical Review Physics Education Research*, 20, 020108.
10. Klaver, B., Rombouts, S. M. A., Fellner, M., Messinger, A., Ender, K., Ludwig, K., & Lechner, W. (2026). SWAP-less implementation of quantum algorithms. *Physical Review A*, 113, 012443. <https://doi.org/10.1103/2wzk-fnhx>
11. Meyer, J. C., Passante, G., Pollock, S. J., & Wilcox, B. R. (2022). Today's interdisciplinary quantum information classroom: Themes from a survey of quantum information science instructors. *Physical Review Physics Education Research*, 18, 010150.
12. Mykhailova, M., & Svore, K. M. (2020). Teaching quantum computing through a practical software-driven approach: Experience report. *arXiv:2010.07729*.
13. Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information*. Cambridge University Press.
14. Trounson, A. (2018). Lifting the lid on quantum computing. Pursuit, University of Melbourne. <https://pursuit.unimelb.edu.au/articles/lifting-the-lid-on-quantum-computing>
15. University of Melbourne. (n.d.-a). Our teaching: Elements of Quantum Computing. <https://www.unimelb.edu.au/cbmm/teaching/overview>
16. University of Melbourne. (n.d.-b). Welcome to QUI; QUI in teaching. <https://qui.research.unimelb.edu.au/>
17. Wootton, J. R., Harkins, F., Bronn, N. T., Carrera Vazquez, A., Phan, A., & Asfaw, A. T. (2020). Teaching quantum computing with an interactive textbook. *arXiv:2012.09629*.
18. Feynman, R. P., Leighton, R. B., & Sands, M. (1965). *The Feynman Lectures on Physics*, Vol. III. Addison-Wesley.
19. IBM Quantum Documentation. (2026). Introduction to transpilation; Transpiler stages. <https://quantum.cloud.ibm.com/docs/en/guides/transpile> and <https://quantum.cloud.ibm.com/docs/en/guides/transpiler-stages>
20. Aumann, P., Fellner, M., Alber, D., Cykiert, M., Fleckenstein, C., ter Hoeven, R., Stenzel, L., Valencia-Tortora, R. J., & Lechner, W. (2026). Demonstrating record fidelity for the quantum Fourier transform. *arXiv:2604.12465*. [Preprint].
21. Ghazi Vakili, M., Gorgulla, C., Snider, J., et al. (2025). Quantum-computing-enhanced algorithm unveils potential KRAS inhibitors. *Nature Biotechnology*, 43, 1954-1959. <https://doi.org/10.1038/s41587-024-02526-3>
22. Google Quantum AI and Collaborators. (2025). Quantum error correction below the surface code threshold. *Nature*, 638, 920-926. <https://doi.org/10.1038/s41586-024-08449-y>
23. Atom Computing and Collaborators. (2026). Quantum error correction with the toric code. *arXiv:2606.04079*. [Preprint].
24. Iqbal, M., Lyons, A., Lo, C. F. B., et al. (2025). Qutrit toric code and parafermions in trapped ions. *Nature Communications*, 16, 6301. <https://doi.org/10.1038/s41467-025-61391-z>

Archive note: contemporary examples were checked against the cited primary publication or preprint as of 31 July 2026. Preprints are labelled explicitly and should be rechecked before future publication.